

The MOND Acceleration Scale from Gas-Dominated Rotation-Curve Points: a Mass-to-Light-Robust Estimate and its Distance Systematic

The authors — 2026-10-05, version 1.0 — AI-assisted research programme; not peer reviewed

Abstract

Published values of the MOND acceleration scale scatter by $\sim 20\%$ because a_0 is degenerate with the stellar mass-to-light ratio Υ : in the deep regime $a_0 \simeq g_{\text{obs}}^2/g_{\text{bar}}$. We remove most of that degeneracy by fitting only rotation-curve *points* at which gas supplies more than 80% of the baryonic acceleration. In SPARC this leaves 124 points in 19 galaxies, all deep in the MOND regime ($y = g_{\text{bar}}/a_0 \approx 0.04$). Their a_0 changes by only 11% as Υ runs from 0.3 to 0.7. They also prefer the quadrature kernel $\nu = \sqrt{1 + 1/y}$ over the exponential RAR form ($\Delta\chi^2 = +5.0$, independent of Υ), which cuts the kernel systematic to $\sim 3\%$. Selecting points rather than galaxies is essential: galaxy-level “gas-rich” samples retain a 63% Υ dependence. The result then depends on distance, because for gas-dominated baryons a_0 scales roughly as D^{-2} to D^{-4} . The eight galaxies with TRGB or Cepheid distances give $a_0 = 1.16 \times 10^{-10} \text{ m s}^{-2}$ ($\pm 20\%$). The eleven with Hubble-flow distances give 0.70×10^{-10} ($\pm 18\%$). WALLABY DR2 (46 galaxies, Hubble flow) gives 0.73×10^{-10} ($\pm 23\%$), and LITTLE THINGS (22 galaxies, authors’ baryon models) 0.78×10^{-10} ($\pm 41\%$). A random-effects combination gives $a_0 = (8.3 \pm 1.1) \times 10^{-11} \text{ m s}^{-2}$ ($I^2 = 24\%$). On it the conventional 1.2×10^{-10} is disfavoured at 2.7σ ; the values tying a_0 to the dark-energy density ($\frac{1}{2}c\sqrt{G\rho_\Lambda} = 0.94$, $cH_\Lambda/6$, $cH_\Lambda/2\pi$) all lie within 0.9σ ; the critical-density value 1.13 is at 2.3σ . That verdict is distance-conditional: three of the four inputs rest on Hubble-flow distances or external baryon models, and the TRGB/Cepheid subset alone favours 1.13. Distinguishing the candidate coefficients needs distances good to $\sim 3\text{--}5\%$ for gas-dominated dwarfs, not more galaxies.

1 Method: gas-dominated points

For each SPARC galaxy [LMS16] (quality ≤ 2 , $i \geq 30^\circ$) we keep the radii where $V_{\text{gas}}^2/V_{\text{bar}}^2 > 0.8$ at $\Upsilon_{\text{disk}} = 0.5$ ($\Upsilon_{\text{bul}} = 0.7$), and fit $g_{\text{obs}} = g_{\text{bar}} \nu(g_{\text{bar}}/a_0)$ in $\log g$ with the measured velocity errors plus $\sigma_{\text{int}} = 0.11$ dex, profiling a_0 . Errors come from a galaxy bootstrap. With a 0.8 cut the result changes by 11.2% between $\Upsilon = 0.3$ and 0.7. A galaxy-level cut (gas fraction > 0.6 at the last point) changes by 63%, because the inner points of such galaxies are star-dominated. On the gas points the exponential RAR kernel is worse than $\sqrt{1 + 1/y}$ by $\Delta\chi^2 = +5.0$ at every Υ , and the allowed family members move a_0 by only $\sim 3\%$.

2 Results

sample (gas-dominated points)	galaxies	points	a_0 ($10^{-10} \text{ m s}^{-2}$)
SPARC, TRGB/Cepheid distances	8	70	$1.158 \pm 20\%$
SPARC, Hubble-flow distances	11	54	$0.697 \pm 18\%$
SPARC, all	19	124	$0.900 \pm 14\%$
WALLABY DR2 (Hubble flow; AllWISE stars)	46	218	$0.728 \pm 23\%$
LITTLE THINGS (authors’ baryon models)	22	720	$0.780 \pm 41\%$
random-effects pool (four independent sets)			$0.832 \pm 13\%$

WALLABY. We used the DR2 kinematic models (rotation curves and face-on H I surface densities) with AllWISE W1 stellar masses. The gas disc was computed by exact ring summation, checked to 0.9% against the Freeman disc. We tested the survey’s assumptions one at a time: the external field of the cluster fields (Hydra, Norma, NGC 4636, NGC 5044), via the quadrature law’s own external-field cubic $g^2 - g_N^2 = a_0 g_N g / (g + \sqrt{2}g_{\text{ext}})$ with per-galaxy g_{ext} , moves a_0 by +1 to +8.5%; the H I flux correction by 15%; asymmetric drift (8 km s^{-1}) by 5%; CMB-frame instead of Hubble-flow distances by -33% . Distances dominate.

LITTLE THINGS. For the three galaxies shared with SPARC (DDO 154, DDO 168, NGC 2366), the two teams’ baryon models give per-galaxy a_0 that differ by a factor 1.6–2.4. Modelling choices, not only statistics, move a_0 by tens of percent.

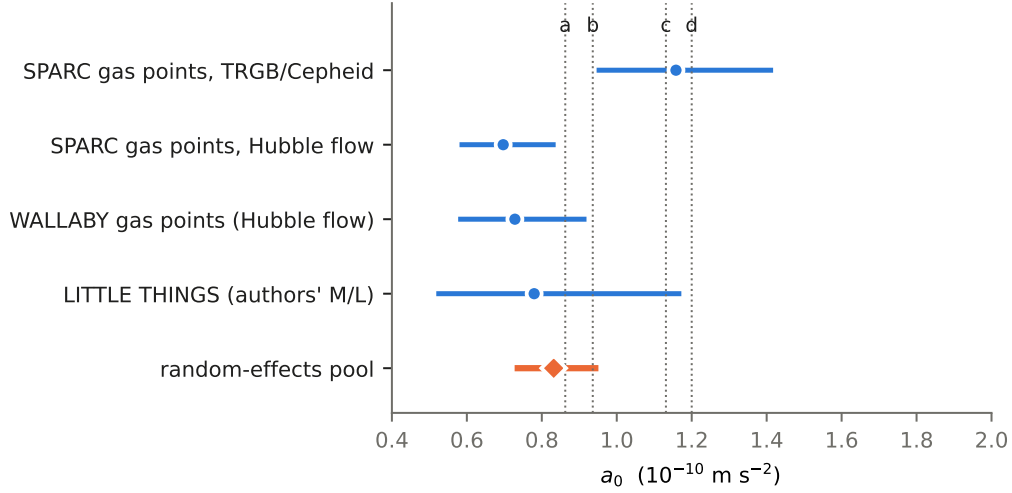


Figure 1: The Υ -robust estimates (circles, 68%) and their random-effects pool (diamond), with candidate values of a_0 (dotted): (a) $cH_\Lambda/2\pi$, (b) $\frac{1}{2}c\sqrt{G\rho_\Lambda}$, (c) $\frac{1}{2}c\sqrt{G\rho_{\text{crit}}}$, (d) the conventional 1.2×10^{-10} .

3 Combination and candidate values

The four sets share no galaxies, except three that LITTLE THINGS shares with SPARC. A DerSimonian–Laird random-effects combination gives $Q = 3.97$ (3 degrees of freedom), $I^2 = 24\%$ and $a_0 = 8.32 \times 10^{-11} \text{ m s}^{-2}$ ($\pm 13\%$). Pulls of candidate values: $\frac{1}{2}c\sqrt{G\rho_\Lambda}$ (9.36) $+0.88\sigma$; $cH_\Lambda/6$ (9.03) $+0.62$; $cH_\Lambda/2\pi$ (8.63) $+0.27$; $cH_0/2\pi$ (10.4) $+1.69$; $\frac{1}{2}c\sqrt{G\rho_{\text{crit}}}$ (11.3) $+2.30$; 1.2×10^{-10} $+2.74$.

The distance caveat is load-bearing. The Hubble-flow subsets sit low and the TRGB/Cepheid subset high (ratio 1.66, $\sim 2\sigma$). On the redshift-independent subset alone, the critical-density value is the closest ($+2\%$). For gas-dominated points the per-galaxy scatter is 0.60 in $\ln a_0$. Separating candidates 4–9% apart at 2σ needs ~ 100 –1000 such galaxies *and* a shared distance-plus-kernel systematic below ~ 3 –5%; today it is $\sim 9\%$. The decisive dataset is therefore one homogeneous analysis of many gas-dominated dwarfs with TRGB distances.

4 Summary

Gas-dominated rotation-curve points give a mass-to-light-robust a_0 , and they prefer the quadrature kernel over the exponential RAR. Combining independent sets gives $a_0 = (8.3 \pm 1.1) \times 10^{-11} \text{ m s}^{-2}$. This is below the conventional 1.2×10^{-10} at 2.7σ , and consistent with a_0 set by the dark-energy density, but the result depends on the distance scale. No coefficient among $\frac{1}{2}$, $1/6$ and $1/2\pi$ (on ρ_Λ) is preferred. The cold mass of cosmology is not addressed here.

Reproducibility. `sonnet55_push/puzzle_32pi/p41b_gas_points_calibration.py`, `p42_kernel_on_gas_points.py`, `p43_wallaby_gas_points.py`, `p44_samplesize_and_bayes.py`, `p46_gas_points_trgb.py`, `p47_little_things_a0.py`, `p48_meta_analysis.py`; fetch logs `data_assembly/wallaby_dr2/`, `data_assembly/little_things_oh2015/`; figure `make_paper43_figures.py`; audit `PAPER43_audit.py`.

References. [LMS16] Lelli, McGaugh & Schombert, AJ 152, 157 (2016) (SPARC). WALLABY DR2: Murugesan et al., arXiv:2409.13130 (2024); kinematic models via CADCE. LITTLE THINGS: Oh et al., AJ 149, 180 (2015), Vizier J/AJ/149/180. AllWISE: IRSA `allwise_p3as_psd`. Random effects: DerSimonian & Laird, Controlled Clinical Trials 7, 177 (1986).